Map Description and Prop Placement

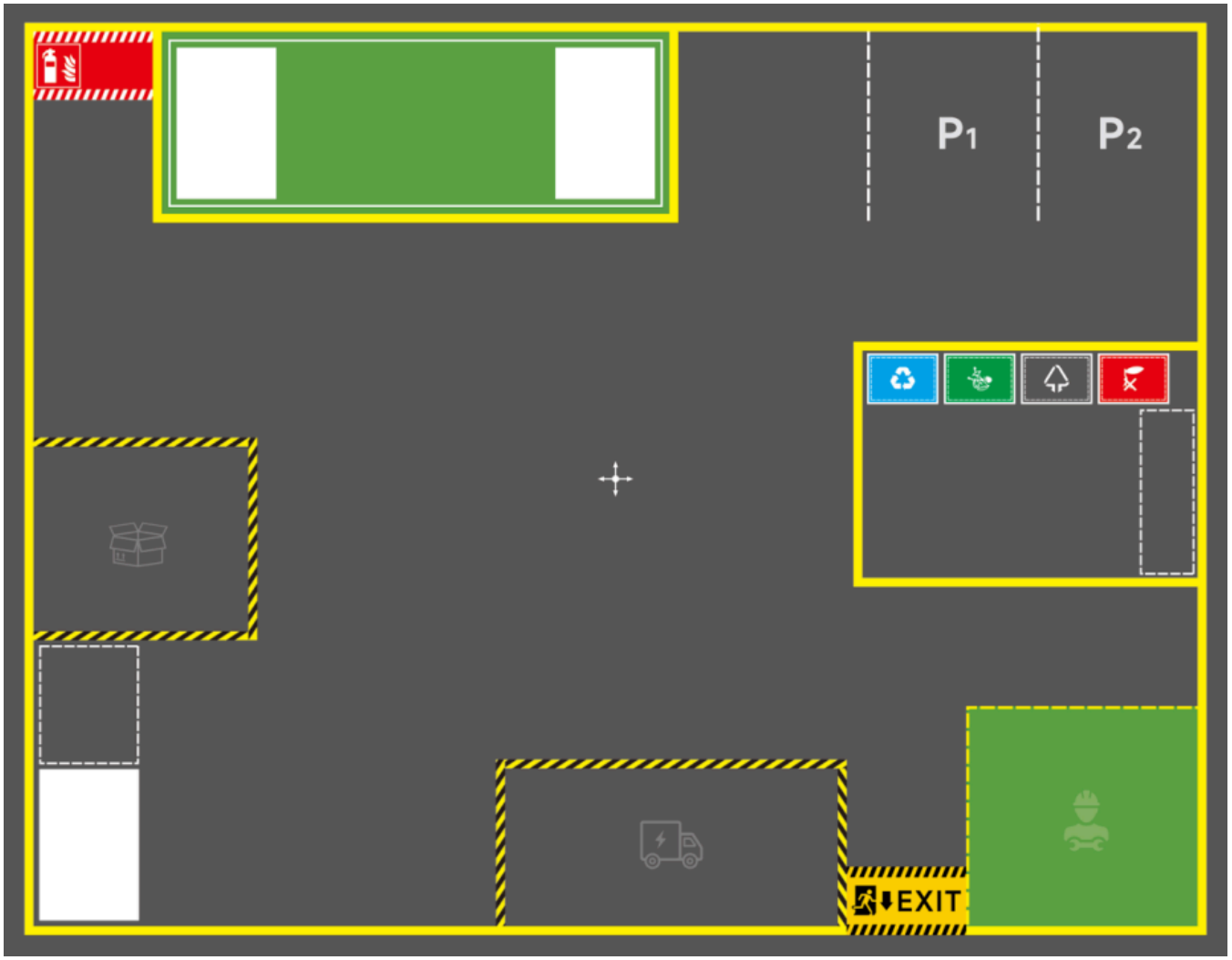
Map Description and Prop Placement

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1. Map Overview

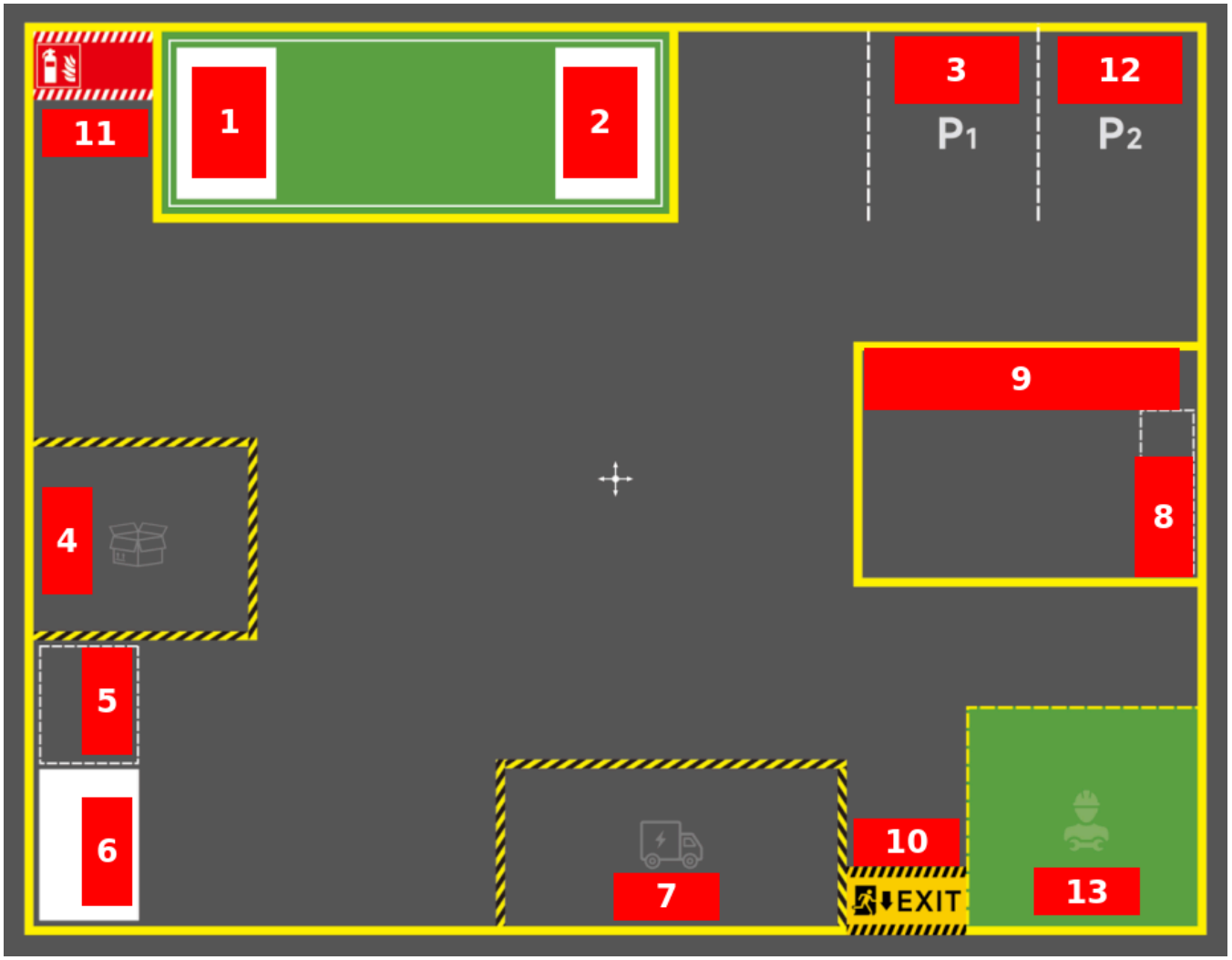
This sandbox map is specifically designed for embodied intelligent mobile manipulation robots - a **"Smart Unmanned Factory and Flexible Logistics Warehouse Miniature Training Scenario"**.

In actual industrial scenarios, composite mobile robots need to shuttle between different workstations such as shelves, material warehouses, assembly production lines, and garbage collection points, performing fully autonomous navigation, delivery, and sorting-grasping operations. This sandbox highly reproduces the real production workflow of modern smart factories through high-definition ground lane lines, functional area markings, and column targets.



2. Detailed Functional Area Division

The naming and functional division of each area in the sandbox supports free definition and expansion. You can customize place names according to your teaching or R&D needs in subsequent courses. The following is the factory-recommended typical smart factory functional layout reference:



Area Function Comparison Table

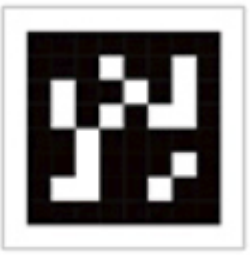
Area No.	Area Name	Corresponding Props and Facilities	Typical Applications and Task Scenarios
1	**Shelf Area 1**	Double-layer storage shelf, material tray, cargo cubes	Raw material outbound, semi-finished product storage, high-precision fixed-point placement
2	**Shelf Area 2**	Double-layer storage shelf, material tray	Finished product inbound, cross-station material transfer destination
3	**Parking Lot 1 (Start/Charging Pile)**	Ground marking parking spaces	Robot initial departure point, inspection end auto-return charging parking

Area No.	Area Name	Corresponding Props and Facilities	Typical Applications and Task Scenarios
4	**Goods Warehouse (Raw Material Reserve)**	Blue anti-static mat, various colored cargo cubes	Basic raw material storage area, on-demand replenishment extraction station
5	**Goods Sorting Area (Flexible Processing Table)**	Blue mat, mixed color cubes	Visual color recognition, multi-category brick autonomous grasping and classification
6	**Shelf Area 3**	Storage shelf, classification tray	Pre-processing buffer area, cross-area production line transfer hub
7	**Shipping Area (Shipping Dock)**	Packing shipping table, goods cubes waiting for shipping	Material factory inspection, active inspection and replenishment of missing goods
8	**Garbage Sorting Area**	Scattered irregular domestic garbage cubes	YOLO-OBB rotating object detection, tilt-adaptive classification grasping
9	**Garbage Bin Placement Area**	Four-color classified garbage bins (food waste/recyclable/other/hazardous)	Stable action group delivery, environmental intelligence classification practice
10	**Factory Entrance/Exit**	Channel gate markings	Cross-workshop main line entrance/exit navigation testing
11	**Fire and Safety Equipment Area**	Emergency supplies marking positions	Factory safety inspection check-in, abnormal obstacle detour testing
12	**Parking Lot 2 (Backup Parking Position)**	Backup parking markings	Multi-vehicle scheduling collaboration, second standby parking point
13	**Equipment Maintenance and Quality Inspection Area**	Maintenance station markings	Equipment health self-check, fault simulation check-in point

3. Experimental Props Function and Technical Description

3.1 AprilTag / ArUco Machine Code Tags

- **Technical Principle**: Black and white high-contrast matrix code with unique digital ID and 4 physical feature corner points.
- **Function Description**: Attached to sandbox shelf columns, key intersections, or operation table edges. When the cart navigates long distances to near a workstation, if slight drift occurs due to wheel slipping, the camera only needs to glance at the tag code, and the algorithm can instantly solve the global 3D coordinates, achieving **centimeter-level visual global repositioning correction**.



3.2 Storage Shelves and Material Trays

- **Physical Parameters**: Adapted to DOFBOT Pro robotic arm vertical lift and horizontal outreach workspace.
- **Function Description**: Simulates industrial stereo shelves. Upper and lower trays are used to place various colored cargo cubes, training the robot's spatial height perception and inverse kinematics (IK) placement capability.



3.3 Blue Cargo Anti-Static Mat

- **Function Description**: Simulates anti-static operation tables and receiving tables in industrial dust-free workshops. The mat surface is flat with soft diffuse reflection characteristics, facilitating the robot's RGB-D depth camera to obtain high-quality surface point cloud data, preventing point cloud missing due to strong light reflection.



3.4 Four-Color Classified Garbage Bins

- **Function Description**: Corresponding to the national standard four-classification garbage bins (green food waste, blue recyclable, gray other waste, red hazardous waste). In garbage sorting tasks, the robotic arm, based on YOLO-OBB detected category labels, calls preset action groups to precisely deliver to the corresponding colored bin opening.



3.5 Various Colored Cargo Cubes and Garbage Model Cubes

- **Color Cubes (Standard 4x4x4 cm wooden cubes)**: Covering red, green, blue, yellow four colors, simulating various industrial standard parts, used for basic tracking grasping, semantic fixed-point placement, and cross-station transfer.
- **Garbage Cubes**: Surface printed with domestic garbage patterns such as peach pits, banana peels, cans, batteries, etc., used for tilt-adaptive rotating object detection and sorting grasping.

